BAA1027 Data Analytics: Machine Learning & Advanced Python

Research Project: Machine Learning Analysis on Company Bankruptcy

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AI Declaration Statement

A. ☐ I did not use any Gen AI tools (including for editing and proofreading) in the completion of this assignment

B. ✓ I did use Gen AI tools in the completion of this assignment

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Date: 19/04/2026

Tools:	Google Gemini 3 Fast and Google Search Gemini AI
Outputs:	Used Gemini 3 Fast for proofreading, pointing out any errors in writing, or pointing out confusing writing etc. Google Search Gemini AI automatically explained concepts and problems I was not familiar with when googled, offering examples, solutions and further reading.
Prompts:	Google Gemini 3 Fast - “Proofread the attached assignment. Point out ANY and ALL grammar or spelling mistakes, or sentences that do not make sense/poorly worded” Google Search Gemini AI - “k-nearest-neighbours” etc.

Abstract

Rising global bankruptcy rates have become a serious issue for the global economy, particularly for investors and financial institutions who are at risk of financial loss. Machine learning is increasingly being used to predict such bankruptcies in order to provide stakeholders with additional information and prevent such potential financial loss. This study utilises four ML models (SVM, RF, KNN, XGBoost) onto a Taiwanese Bankruptcy dataset from Kaggle. Using SMOTE, SelectKBest feature selection and F2/AUC-ROC as primary metrics, it evaluates the best possible ML models for bankruptcy prediction. XGBoost achieved the best F2 of 0.554, with SVM achieving the highest recall at 0.841. All models exceeded a benchmark F-2 score of 0.423 from recent literature using the same dataset. SVM is recommended as the best tool for predicting bankruptcies, with XGBoost recommended as the best balance between predicting bankruptcies and controlling false alarms.

I. Introduction

Following the COVID-19 pandemic which saw the world economy grind to a halt, bankruptcy has become an endemic issue within the world economy. Previous studies have shown that there was a 17% increase in bankruptcies in 2024 compared to 2023, despite governments globally injecting an estimated $9 trillion to keep businesses afloat (Iparraguirre-Villanueva and Cabanillas-Carbonell, 2024). As of early 2026, this trend has continued unabated, with US bankruptcy filings increasing 14% in Q1 of the year (Bashir, 2026). As a result, it has become increasingly essential for lenders, investors and government regulators to be able to detect and respond to potential business bankruptcies to prevent significant loss of capital and to avoid harm to domestic economies. The potential solution to this issue can be traced back to 1966, in which (Beaver, 1966) applied univariate analysis to financial ratios in order to detect bankruptcy. This was followed up by (Altman, 1968), who created "Altman's Z-score" using Multiple Discriminant Analysis (MDA) analysis to do the same thing, utilising fewer financial ratios and statistics.

Utilising modern machine learning approaches on a Taiwanese Bankruptcy dataset from Kaggle ("Taiwanese Bankruptcy Prediction," 2020), with data from 1999 to 2009, this paper seeks to address the large class imbalance problem which has long degraded performance in bankruptcy prediction utilising ML methods in literature. Due to the rarity of bankruptcy, datasets in current literature tend to have extremely small bankrupt classes (~3%), leading to artificially high accuracy. To do so, "Support Vector Machine", "Random Forest", "K-Nearest Neighbour" and "Xtreme Gradient Boost" models, in tandem with SMOTE oversampling techniques to overcome imbalance issues and feature selection to reduce computational intensity will be used to examine the data. Additionally, AUC-ROC and a modified F-2 score will be used as the primary evaluation metrics to accurately examine the true accuracy and applicability of the models alongside Recall and Precision.

II. Related Works

The literature on bankruptcy prediction using statistical and machine learning methods can be traced back originally to 1966, with Beaver's univariate analysis of 30 different financial ratios to predict bankruptcy in firms (Beaver, 1966). In 1968, this was followed up by Altman who utilised MDA with 5 financial ratios to do the same, which could perhaps be considered an early implementation of feature selection. Since then, ML tools have become a popular alternative method of predicting bankruptcy, eventually utilised by Altman himself in a follow up study. In this study, Altman's Z-Score was compared with modern ML methods. There was a particular emphasis on ensemble methods in this follow up paper, which would go on to inspire the use of XGBoost in this study. It was noted that machine learning methods have a roughly 10% improvement over traditional models, such as Altman's Z-score (Barboza, Kimura and Altman, 2017). It should be noted that neither feature selection nor transformation was used on a dataset with over 10,000 observations in this study, which may have handicapped the ML models used.

The dataset for this study has also been used by two other separate studies, first by (Liang *et al.*, 2016) who used it to measure not only the impact of financial ratios, but corporate governance indicators in bankruptcy prediction. It should be noted that the corporate governance indicators are not present in this dataset used for this study, but otherwise the datasets are identical. Liang et al. addressed the large class imbalance problem through stratified sampling to create a dataset of 239 bankrupt, and 239 non-bankrupt firms, balancing the classes. Additionally, 10-fold cross-validation was utilised to avoid a variability of samples. The 10-fold cross-validation technique was adopted to be utilised in this study. This dataset was also used by (Wang and Liu, 2021), who attempted to address the imbalance in classes through Edited Nearest Neighbors, an undersampling technique. It was noted by

Wang and Liu that oversampling has, thus far, been the dominant method for addressing the class imbalance issue in bankruptcy detection literature. This served as the primary inspiration for utilising SMOTE in this study, to compare the possible differing results of implementing oversampling as opposed to undersampling. Wang and Liu additionally utilised F-2 scores as opposed to traditional F-1 scores in order to double the emphasis placed on Recall, with the goal of emphasising the ML model focus on identifying bankruptcies even at the sake of higher false alarms. Considering this to be logical considering the higher financial consequences of missing a bankruptcy as opposed to falsely labeling a business as potentially bankrupt, this too was adopted to be used in this study.

SMOTE techniques have also been used in literature such as (Narvekar *et al.*, 2021). Sharing many of the models used in the present study, this paper used RF, SVM and XGBoost, but noticeably had a 99% accuracy rate, with a 99% AUC-ROC. Compared to the consistently lower rates of accuracy in literature, it appears likely that this was the result of a data leak within their models. As such, it provided inspiration to examine these three models again but to avoid any potential data leak.

Regarding feature selection, (Du *et al.*, 2020) proposed that in order to tackle the "Curse of Dimensionality" and associated computational costs, feature selection is required, which provided the inspiration to only examine the 20 most correlated variables with bankruptcy in this study.

Finally, regarding the models themselves, there has been consistent debate surrounding the "best" ML model in bankruptcy prediction. However, SVM, RF, KNN and XGBoost have been consistently popular to examine. Their success has been mixed, with SVM performing best in (Liang *et al.*, 2016) but poorly in (Wang and Liu, 2021). RF has dominated in other

studies such as (Lombardo *et al.*, 2022) and (Islam *et al.*, 2024). XGBoost has also performed well in (Shetty, Musa and Brédart, 2022). As such, these studies and their conflicting results all encouraged further examination of these models.

III. Methodology

Before delving into the methodology of this study, it is first important to understand the dataset itself. It contains 6,819 companies, 95 financial ratio features, 220 bankrupt companies (3.2%) and 6,599 non-bankrupt companies. No null values are present, and no duplicates were found ("Taiwanese Bankruptcy Prediction," 2020). There is a clear data imbalance present, which has been tackled utilising SMOTE.

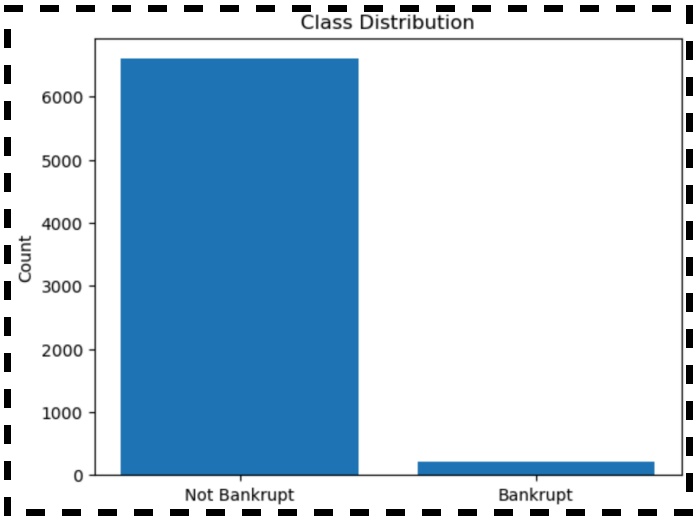

Figure 1 - Class Distribution

SMOTE has been utilised to balance the dataset to have 5279 bankrupt companies and 5279 non-bankrupt companies though oversampling. This is done by SMOTE through creating synthetic samples of data drawn along the lines between minority class K-nearest neighbours (Narvekar *et al.*, 2021). However, it is important to note that this is not applied to the test

data, in order to preserve real world class distribution. It is solely for the purpose of training the models.

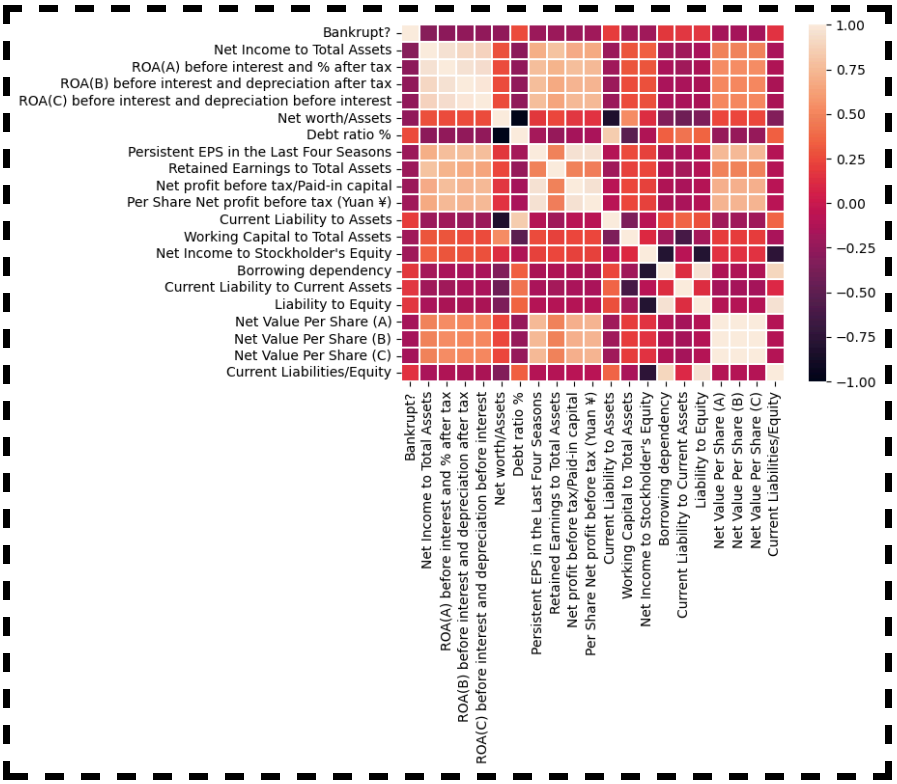

Figure 2 - Correlation Heatmap

Brief exploratory data analysis was also performed by producing a correlation heatmap of the top 20 correlated features with bankruptcy. It revealed that features relating to assets and liabilities primarily dominate correlation with bankruptcy.

Literature varied on the best train-test split for ML models, with (Narvekar *et al.*, 2021) and (Antulov-Fantulin, Lagravinese and Resce, 2021) implementing 70-30 splits, and (Iparraguirre-Villanueva and Cabanillas-Carbonell, 2024) utilising 80-20 splits. Ultimately, it was decided to implement an 80-20 stratified split in order to keep the ratio of each class

equal in each split. This in conjunction with SMOTE was intended to ensure that the class imbalance of our dataset did not negatively affect model performance.

In order to reduce computational intensity on the limited resources available for this study, feature selection was implemented using SelectKBest, with k=20 to reduce noise and computational cost. Whilst potentially losing some valuable information, this is consistent in literature such as in (Chen, 2011) and (Du *et al.*, 2020) in order to reduce computational intensiveness, although methods differ, such as with PCA implementation in other studies.

Four different ML models were used in this study. SVM, RF, KNN and XGBoost. SVM is a supervised ML method primarily used for classifications, and is a class of optimal margin classifiers that finds the boundary hyperplane with maximal margin between clusters of different classes (Shetty, Musa and Brédart, 2022). It has been used consistently in literature, and in most of the studies mentioned thus far.

Random Forest is based on decision tree models. Similar to bagging, it repeatedly generates classification functions based on subsets, building multiple decision trees and deciding on classification based on majority vote. It is particularly useful for avoiding overfitting (Barboza, Kimura and Altman, 2017). In this study, 100 decision trees were utilised.

KNN is a more simple machine learning technique, which simply bases classification on its K-nearest neighbours. K=5 was used in this study.

XGBoost is a boosting ensemble method, which takes a set of easily learnable weak classifiers and turns them into a more powerful classifier by having each model correct the errors of the previous one. XgBoost is a variant of gradient boosting, and uses a more regularised method to control overfitting (Shetty, Musa and Brédart, 2022).

With regards to evaluation, the primary tools for evaluation in this study are F-2 score (a re-weighted F–1 score, with recall weighted twice as much, minimising false negatives e.g not identifying a bankruptcy), AUC-ROC, which was used to prove the accuracy of the model (in which it must score over 0.5 to be better than random chance), alongside a confusion matrix to provide an easy visual guide of performance. 10-Fold cross validation was also used to measure training performance to ensure evaluation accuracy, such as in (Liang *et al.*, 2016).

IV. Results

```
  SUMMARY
--------------------------------------------------
               CV F2    AUC  Recall  F2 (Bankrupt)  FN (missed)
SVM            0.902  0.911   0.841          0.474          7.0
Random Forest  0.982  0.922   0.614          0.517         17.0
KNN            0.978  0.815   0.659          0.428         15.0
XGBoost        0.984  0.908   0.682          0.554         14.0

Best Recall : SVM (0.841)
Best F2 (Bankrupt) : XGBoost (0.554)
Best AUC : Random Forest (0.922)
Best CV F2 : XGBoost (0.984)
```

Figure 3 - Results Summary

XGBoost

Overall, based on the utilised evaluation measures, XGBoost shows the most impressive result. The main metric of F-2 stands at 0.554, higher than any other model. In other words, when weighing the recall of the bankrupt class twice as heavily, XGBoost has the highest overall accuracy. Additionally, with a recall of 0.682, detecting 30 out of 44 bankruptcies, it stands at second overall. Interestingly, it performs rather poorly in AUC, in third place but

still within the “excellent” range of 0.8 to 0.9 described in (Islam *et al.*, 2024). It should also be considered that XGBoost requires relatively low computational power, at least in comparison to SVM.

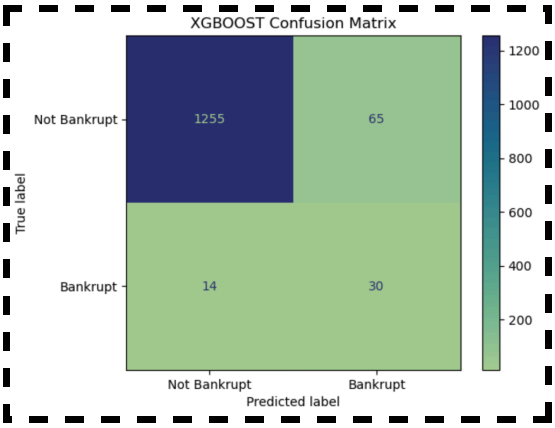

Figure 4 - XGBoost Confusion Matrix

```
XGBOOST
--------------------------------------------------
Cross-validation scores: [0.98311445 0.98428144 0.98644578 0.98053892 0.9838528  0.98985725
 0.98050244 0.98379193 0.99022924 0.9762533 ]
Mean cross-validation score: 0.9838867542864863
              precision    recall  f1-score   support

Not Bankrupt       0.99      0.95      0.97      1320
    Bankrupt       0.32      0.68      0.43        44

    accuracy                           0.94      1364
   macro avg       0.65      0.82      0.70      1364
weighted avg       0.97      0.94      0.95      1364

AUC: 0.9077823691460055
F2 Score (Bankrupt): 0.5535055350553506
Confusion Matrix:
Recall: 0.6818181818181818
```

Figure 5 - XGBoost Summary

SVM

Following behind XGBoost, SVM comes in at an arguable second place, despite a poor F-2 score of 0.474, the second lowest. This is due to its extremely high recall of 0.841, catching 37 out of 44 bankruptcies, alongside an AUC of 0.910, above XGBoost. Its poor F-2 score is a result of a lacking precision of 0.17, flagging many healthy companies as bankrupt with a very high false alarm rate.

However, it should be acknowledged that SVM is arguably the most useful for real life application by creditors and investors. The cost of a missed bankruptcy is far more catastrophic than a false alarm, though certainly burdensome.

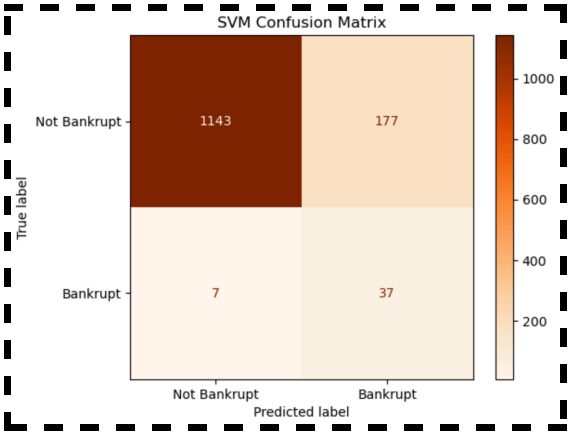

Figure 6 - SVM Confusion Matrix

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SVM
--------------------------------------------------
Cross-validation scores: [0.89418778 0.90808006 0.90363705 0.91420007 0.89824823 0.881
 0.89906542 0.90704648 0.90314136 0.90601504]
Mean cross-validation score: 0.9015195453781913
              precision    recall  f1-score   support

Not Bankrupt       0.99      0.87      0.93      1320
    Bankrupt       0.17      0.84      0.29        44

    accuracy                           0.87      1364
   macro avg       0.58      0.85      0.61      1364
weighted avg       0.97      0.87      0.90      1364

AUC: 0.9109676308539945
F2 Score (Bankrupt): 0.47435897435897434
Confusion Matrix:
Recall: 0.8409090909090909
```

Figure 7 - SVM Results Summary

Random Forest

Random Forest produced the best AUC with a score of 0.922, the only model in (Islam *et al.*, 2024)'s "exemplary" range of being above 0.9 comfortably. It had the second best F-2 score at 0.517, detecting 24 out of 44 bankruptcies. However, it had the lowest recall of any model, which it was making up for with precision, a clear example of the precision-recall tradeoff. Considering the core objective of catching potential bankruptcies, this is a clear weakness.

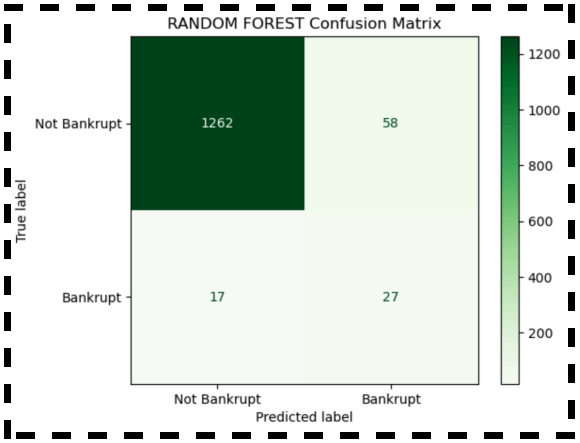

Figure 8 - Random Forest Confusion Matrix

```
RANDOM FOREST
--------------------------------------------------
Cross-validation scores: [0.97972973 0.98573038 0.98305085 0.97597598 0.97965335 0.98382242
 0.97895528 0.98679245 0.98718915 0.98187311]
Mean cross-validation score: 0.9822772701775724
              precision    recall  f1-score   support

Not Bankrupt       0.99      0.96      0.97      1320
    Bankrupt       0.32      0.61      0.42        44

    accuracy                           0.95      1364
   macro avg       0.65      0.78      0.69      1364
weighted avg       0.97      0.95      0.95      1364

AUC: 0.9217716942148761
F2 Score (Bankrupt): 0.5172413793103449
Confusion Matrix:
Recall: 0.6136363636363636
```

Figure 9 - Random Forest Results Summary

KNN

The undeniable weakest performer overall, with the lowest AUC of 0.815 and lowest F2 of 0.427, is KNN. Its low AUC is the most notable element of this model, which can be seen in Figure 12.

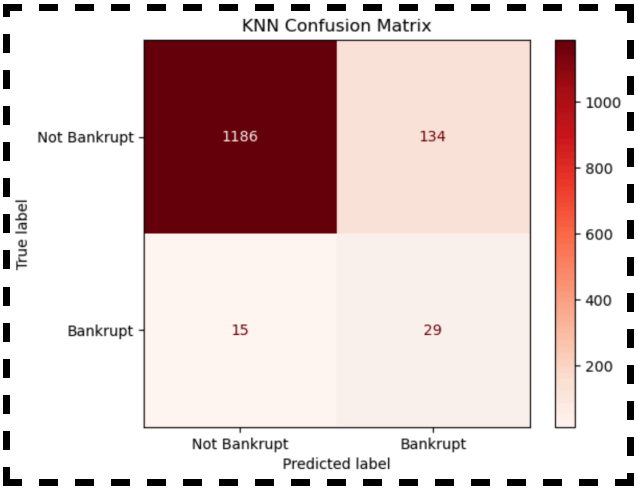

Figure 10 - KNN Confusion Matrix

```
KNN
--------------------------------------------------
Cross-validation scores: [0.97633136 0.97995546 0.98101266 0.97037037 0.97402597 0.9795539
 0.97777778 0.98394324 0.97951583 0.97882615]
Mean cross-validation score: 0.9781312722727409
              precision    recall  f1-score   support

Not Bankrupt       0.99      0.90      0.94      1320
    Bankrupt       0.18      0.66      0.28        44

    accuracy                           0.89      1364
   macro avg       0.58      0.78      0.61      1364
weighted avg       0.96      0.89      0.92      1364

AUC: 0.8146091597796145
F2 Score (Bankrupt): 0.4277286135693215
Confusion Matrix:
Recall: 0.6590909090909091
```

Figure 11 - KNN Results Summary

ROC Curve

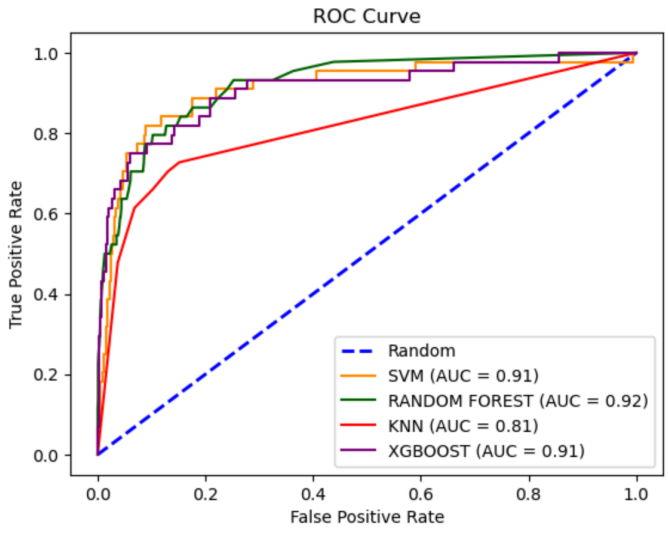

Figure 12 - ROC Curve

In Figure 12, our ROC curve, it is immediately noticeable that all models substantially outperform the random baseline, confirming that the ML analysis in this study is meaningful. That being said, KNN visibly lags behind, particularly in the 0.1 to 0.3 FPR range. Other than this, it is clear that SVM, RF and XGBoost share a very similar and impressive ROC.

It should be noted that a considerable gap exists between cross-validation F-2 scores and F-2 scores across all models, mostly notably XGBoost, with a CV F-2 of 0.984 against a test result of 0.554. This is likely as a result of either SMOTE leakage within training (which critically was not applied to the test data, leaving them as a valid reflection of real world performance) or a sign that the models were overfitting to the balanced distribution of a 50-50 class dataset, and were not able to work as efficiently on a 97-3 class dataset.

Comparison with Other Literature

Perhaps the most interesting comparison to be drawn from available literature is with (Wang and Liu, 2021), who, in using F-2 as an evaluation measurement on an identical dataset, achieved a best F-2 score of 0.423 using Naive Bayes and ENN undersampling. All four of the models in this study exceed this score, with XGBoost surpassing it by 0.13, at 0.554. Particularly interesting is that the XGBoost model used in Wang and Liu's study had a mean F-2 score of 0.295, suggesting that the strength of this study may be its usage of oversampling as opposed to undersampling, or through feature selection, which was not performed in their study. Also notable is that all AUC scores exceed 0.8, meeting (Islam *et al.*, 2024)'s "excellent" threshold.

Also worth noting is that this study has struggled with the precision-recall trade-off in a similar fashion to (Lombardo *et al.*, 2022), where all of the best performing models in said study exhibited a high recall but low precision for the bankrupt class. Lombardo similarly argues that a higher error rate is acceptable with regards to precision, if it means being able to successfully detect bankrupt companies. This nonetheless highlights how this is a problem endemic to bankruptcy detection.

V. Conclusion

This study has shown that it is exceedingly difficult to identify one "best" model when examining through various evaluation measures. Particularly in bankruptcy detection, it may be up to individual investors/credit institutions whether they are willing to tolerate a higher false alarm rate to detect bankruptcies, or whether they are willing to take on additional risk to increase their investment options. Depending on one's preference in this case, either SVM

or XGBoost provide solid prediction capabilities. That being said, SVM was by far the most computationally expensive model, which may give XGBoost an edge.

There are a number of clear limitations to this study which should be addressed in the future. Mainly, the extremely small bankrupt class within the used dataset is a limitation. Whilst SMOTE was certainly able to alleviate this to a certain extent, much more success would likely be found in a more balanced dataset. The fact that the dataset is Taiwan specific and pre-2010 also limits its current applicability to the modern day. The extremely strict feature selection of k=20 may have also removed useful features, regardless of its necessity to tackle computational limitations.

Future studies should aim to overcome these issues by gathering more modern, balanced data, and by processing it on more powerful hardware if possible.

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